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AgriHome

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1 PRODUCT CONCEPT

This section describes the purpose, intended use, and target users of the AgriHome product. AgriHome is a smart indoor plant monitoring and early disease-detection system designed to help users observe plant health before visible damage becomes severe.

The system combines repeatable image capture, a rack-mounted camera carriage, backend storage and analysis, optional computer-vision support, and a user-facing Vision Console to provide health insights, warnings, and historical tracking for home, educational, and small-scale agricultural settings.

Research in image-based plant disease detection, agricultural deep learning, plant-image datasets, and plant phenotyping supports the value of consistent image capture and clear decision-support outputs [1, 2, 3, 4].

1.1 PURPOSE AND USE

AgriHome is intended to support plant owners and small-grow operators by monitoring plant condition and identifying potential health concerns as early as possible.

The product focuses on indoor rack or shelf use and is especially suited for small garden plants, student research plants, and other plants that benefit from regular visual observation.

The current AgriHome concept uses a camera attached to a shelf-mounted XY motion system. The camera carriage moves horizontally along the rack and vertically to the target row, then captures an individual plant image at a predefined capture pose.

The Raspberry Pi-class edge device coordinates motion, camera capture, status reporting, and image transfer. The application stores captured images and associated plant records so the Vision Console can show plant history, device status, and optional plant-health analysis results.

In normal use, an operator places plants in the rack, powers the capture hardware, and uses the Vision Console to manage plants, trays, capture schedules, and device status.

AgriHome performs scheduled or on-demand capture without requiring the operator to manually photograph every plant. When the system records a potentially abnormal condition or receives a model-generated warning, it presents the information as decision support so the operator can inspect the plant and decide on next steps.

AgriHome is not intended to physically treat the plant, apply chemicals, water the plant, or replace professional agricultural diagnosis. It is a monitoring, image-history, and decision-support product.

1.2 INTENDED AUDIENCE

The primary intended audience for AgriHome includes home gardeners, apartment growers, student researchers, hobby growers, and educational agricultural programs that need a practical way to monitor plants with less manual effort.

These users may not have advanced agricultural knowledge, but they can benefit from timely plant imagery, organized history, and clear status indicators.

A secondary audience includes senior-design evaluators, project sponsors, controlled-environment researchers, and small greenhouse operators who may use AgriHome as a prototype platform for repeatable plant imaging and monitoring experiments.

IoT and edge-computing research supports the use of connected sensing, distributed devices, local processing, image capture, and data-driven monitoring in controlled growing environments [5, 6].

If made commercially available, AgriHome would be most attractive to users interested in smart gardening, indoor growing, plant-health history, and low-cost agricultural monitoring.

The product is designed as a standalone monitoring system, but it could also serve as a subsystem within a larger smart agriculture or smart home platform.

2 PRODUCT DESCRIPTION

This section provides an overview of the smart plant monitoring system, including its primary features, functions, external inputs and outputs, and product interfaces.

The system is designed to capture individual plant images with a rack-mounted camera carriage and provide actionable information through a Vision Console.

From the perspective of end users, the system enables repeatable observation and feedback. For maintainers and developers, it provides structured capture records, edge-device behavior, and storage references for later review and analysis.

2.1 FEATURES & FUNCTIONS

The AgriHome system is designed to capture plant images, process or store this information, and provide users with plant-health insights.

The system consists of the following core components:

- **Plant rack and camera carriage:** Holds plants and positions the camera through XY movement so each plant can be captured individually.
- **Raspberry Pi edge device:** Coordinates camera capture, movement commands, heartbeat reporting, status updates, and image transfer.
- **Application services:** Accept device communication, store plant and capture records, manage image metadata, and support authenticated Vision Console workflows.
- **Vision Console:** Displays plants, trays, device status, capture history, and optional plant-health results.
- **Optional computer-vision service:** Processes captured images and reports model confidence or health categories when a model is available.

The system performs the following functions:

- Captures individual plant images from predefined rack positions.
- Supports on-demand and scheduled capture workflows.
- Reports edge-device online status and heartbeat information.
- Stores captured images, image metadata, plant records, and related history.
- Displays plant status, capture history, device status, and optional analysis results to the operator.

The system does not perform the following:

- It does not water, spray, fertilize, prune, or physically treat plants.
- It does not guarantee agricultural diagnosis or disease confirmation.
- It does not operate as an outdoor industrial greenhouse controller.
- It does not move the rack or plants; only the camera carriage moves.

2.2 EXTERNAL INPUTS & OUTPUTS

The system receives input from authorized operators, the rack-mounted camera hardware, the Raspberry Pi edge device, and optional analysis services. Primary inputs include operator login credentials, plant and tray metadata, capture commands, capture schedules, target capture poses, device registration data, heartbeat reports, camera images, image metadata, and optional computer-vision output. Primary outputs include Vision Console pages, plant and tray status views, capture-history pages, image previews, stale-device warnings, capture success or failure messages, plant-health status labels, optional model confidence values, and stored image records for later review.

2.3 PRODUCT INTERFACES

The system provides user-facing, hardware, software, and storage interfaces.

User interface:

- Displays plant health status, plant images, trays, and capture history.
- Provides device-management views for registration, heartbeat status, linked trays, capture controls, and schedules.
- Supports operator workflows for reviewing images, requesting captures, and identifying stale or offline devices.

Hardware interface:

- Camera module attached to the rack-mounted XY carriage.
- Raspberry Pi-class edge device connected to the camera and motion hardware.
- Motion components for X-axis and Y-axis travel, including safe travel limits and wiring.

Software interface:

- Authenticated application routes for operator workflows.
- Device-facing API behavior for registration, heartbeat, command polling, image ingest, and status reporting.
- Storage references for captured image files and metadata records.
- Optional computer-vision service interface for plant-health classification or confidence reporting.



Figure 1: Plant rack capture concept used to communicate the physical monitoring arrangement

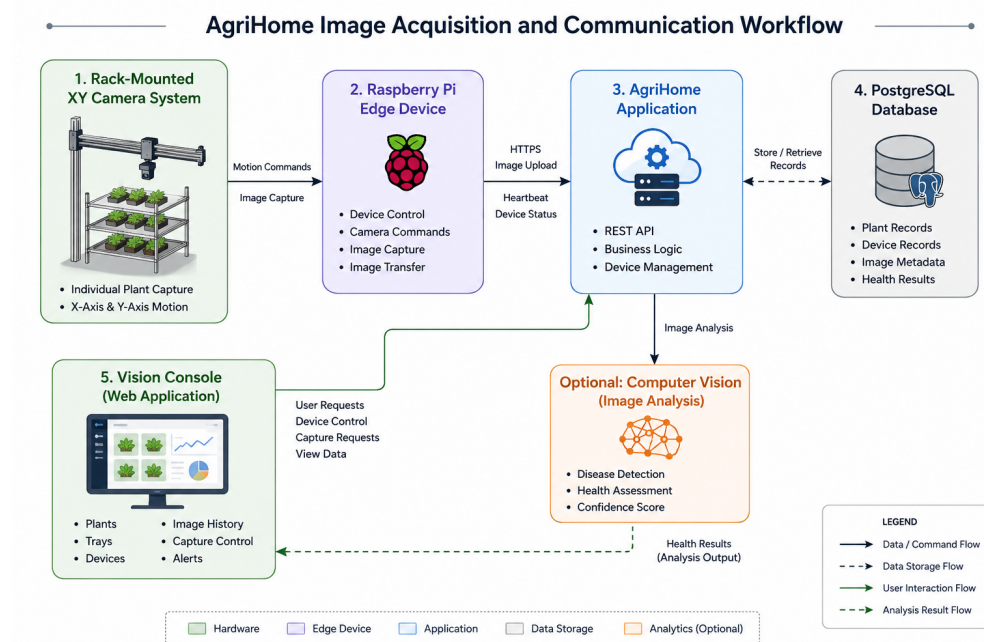


Figure 2: Image acquisition and communication workflow

3 CUSTOMER REQUIREMENTS

This section defines customer requirements for AgriHome. Customer requirements are visible features and functions that intended users, sponsors, and evaluators expect the system to provide.

Each requirement follows the attached SRS template format with Description, Source, Constraints, Standards, and Priority entries.

3.1 PLANT HEALTH STATUS DISPLAY

3.1.1 DESCRIPTION

The system shall provide a user-facing dashboard that displays the current health status of monitored plants and trays.

At minimum, the dashboard shall show a health result such as healthy, warning, or critical, along with the latest available monitoring result or image record for the user to review.

3.1.2 SOURCE

Project scope, sponsor expectations, product description, and plant-disease detection research [1, 4].

3.1.3 CONSTRAINTS

The dashboard must remain simple enough for non-technical users while still showing meaningful monitoring results.

Health status depends on available images, human-entered observations, and optional model-generated results.

3.1.4 STANDARDS

Common dashboard usability, readability, and accessibility practices for browser-based software systems.

3.1.5 PRIORITY

Critical

3.2 AUTOMATIC IMAGE CAPTURE

3.2.1 DESCRIPTION

The system shall capture plant images automatically at configured intervals when a capture schedule is enabled.

The camera shall move to defined rack positions and capture individual plant images rather than relying on one wide contextual image of the entire rack.

3.2.2 SOURCE

Product concept, physical rack design, operator monitoring workflow, and plant imaging research [3].

3.2.3 CONSTRAINTS

Image capture reliability depends on camera hardware, lighting conditions, plant position, rack alignment, XY carriage motion, available storage, network connectivity, and device power.

3.2.4 STANDARDS

Camera operation should follow selected hardware specifications, and captured image records should include timestamps and traceable plant or tray context.

3.2.5 PRIORITY

Critical

3.3 ENVIRONMENTAL DATA COLLECTION

3.3.1 DESCRIPTION

The system should support environmental data collection when approved sensors are available.

Possible measurements include temperature, humidity, soil moisture, or light level associated with plant monitoring.

3.3.2 SOURCE

Product-expansion goals, controlled-environment monitoring practices, and IoT system concepts [5].

3.3.3 CONSTRAINTS

This requirement depends on successful sensor availability, calibration, and communication with the processing unit.

The current buyer-facing prototype does not depend on environmental sensors for its core image-capture workflow.

3.3.4 STANDARDS

Sensor components should follow manufacturer specifications and standard embedded-sensor interfacing practices.

3.3.5 PRIORITY

Medium

3.4 ALERT DISPLAY

3.4.1 DESCRIPTION

The system shall display alerts or notifications when abnormal plant conditions, failed captures, stale device status, or significant monitoring issues are detected.

Alerts shall be visible through the Vision Console and shall indicate that the plant, device, or capture workflow

may require attention.

3.4.2 SOURCE

Product description, customer need, sponsor expectations, device-monitoring workflow, and image-based disease detection research [1, 4].

3.4.3 CONSTRAINTS

Alert usefulness depends on image quality, status-report accuracy, analysis results, and freshness of heartbeat data.

Excessive false alerts may reduce user trust in the system.

3.4.4 STANDARDS

The alert system should follow general software notification practices for clear wording, visible warning presentation, and understandable severity levels.

3.4.5 PRIORITY

Critical

3.5 HISTORICAL DATA DISPLAY

3.5.1 DESCRIPTION

The system shall allow the user to review historical monitoring data, including past captured images, capture timestamps, plant records, health observations, and optional analysis results.

This feature shall support tracking changes in plant condition over time.

3.5.2 SOURCE

Product description, intended user need for traceable plant history, and repeatable plant imaging practices [3].

3.5.3 CONSTRAINTS

History depth may be limited by storage capacity, image size, capture frequency, retention policy, and database design choices within the prototype.

3.5.4 STANDARDS

Stored monitoring data should follow consistent timestamping, image metadata, and record-linkage practices.

3.5.5 PRIORITY

High

3.6 USER SETTINGS INPUT

3.6.1 DESCRIPTION

The system shall accept authorized user input for configuration settings such as capture schedules, rack or tray metadata, plant names, and related behavior settings.

This requirement allows the user to adjust the system to match the monitored rack and plant layout.

3.6.2 SOURCE

Product description, operator workflow, and device-management requirements.

3.6.3 CONSTRAINTS

The number of configurable options may be limited by implementation time, user-interface simplicity, backend support, and validation rules.

3.6.4 STANDARDS

User configuration features should follow common software input validation, authenticated update, and settings-management practices.

3.6.5 PRIORITY

High

3.7 DASHBOARD INTERFACE

3.7.1 DESCRIPTION

The system shall provide a dashboard-based user interface through which users can view plant health status, trays, plants, captures, alerts, device status, and historical monitoring information.

This dashboard shall serve as the primary interaction point between the user and the system.

3.7.2 SOURCE

Product description and project design goals.

3.7.3 CONSTRAINTS

Interface complexity must remain understandable to non-expert users.

Time and frontend development resources may limit visual polish in the prototype, but the workflow must remain usable.

3.7.4 STANDARDS

The interface should follow standard usability, readability, responsive-design, and accessibility practices for browser-based dashboard systems.

3.7.5 PRIORITY

Critical

3.8 NO PHYSICAL PLANT ACTUATION

3.8.1 DESCRIPTION

The system shall not physically interact with or modify the plant.

The product shall not perform watering, spraying, fertilizing, pruning, or any automatic physical treatment.

The moving mechanism is limited to positioning the camera carriage for image capture.

3.8.2 SOURCE

Project scope definition and buyer-facing product concept.

3.8.3 CONSTRAINTS

This requirement limits the system to monitoring, camera positioning, data storage, optional analysis, and display functions.

Any future physical automation would require additional controls, safety requirements, and sponsor approval.

3.8.4 STANDARDS

The requirement follows the documented functional scope of the prototype system.

3.8.5 PRIORITY

Critical

3.9 REMOTE NOTIFICATION SUPPORT

3.9.1 DESCRIPTION

In a future version, the system should support remote notification delivery, such as email, mobile push notification, or text message when abnormal plant conditions or offline devices are detected.

This would allow users to receive alerts even when not actively viewing the Vision Console.

3.9.2 SOURCE

Future customer convenience needs and product-expansion goals.

3.9.3 CONSTRAINTS

This feature would require additional backend services, notification platform integration, secure communication, and user account management.

It may not be feasible within the current prototype scope.

3.9.4 STANDARDS

Future implementations should follow secure communication, opt-in notification, and authentication practices.

3.9.5 PRIORITY

Future

3.10 EDGE DEVICE REGISTRATION

3.10.1 DESCRIPTION

The system shall allow a Raspberry Pi-class edge device to register with the application so that the Vision Console can identify the device, associate it with a tray or rack, and display it to authorized operators.

3.10.2 SOURCE

Edge-device integration workflow, device-management view, IoT architecture, and edge-computing concepts [5, 6].

3.10.3 CONSTRAINTS

Registration must distinguish devices reliably enough for the prototype.

Duplicate registration attempts, missing device identifiers, or network interruptions may require manual operator review.

3.10.4 STANDARDS

Device registration should follow common API practices for unique identifiers, authenticated requests, and traceable device records.

3.10.5 PRIORITY

Critical

3.11 DEVICE HEARTBEAT AND ONLINE STATUS

3.11.1 DESCRIPTION

The system shall receive periodic heartbeat reports from registered edge devices and display online, stale, or offline status in the Vision Console.

3.11.2 SOURCE

Operator reliability workflow, edge-device integration needs, and distributed IoT monitoring concepts [5, 6].

3.11.3 CONSTRAINTS

Network connectivity, device power, API availability, and clock differences may affect status freshness.

The application must avoid presenting stale status as current.

3.11.4 STANDARDS

Heartbeat status should use timestamped records and clear status categories understandable to operators.

3.11.5 PRIORITY

Critical

3.12 SECURE IMAGE INGEST

3.12.1 DESCRIPTION

The system shall accept captured image uploads from authorized edge devices, store the image file or storage reference, and create an associated capture record with plant, tray, timestamp, and device metadata when available.

3.12.2 SOURCE

Capture workflow, image-history requirement, edge-device integration, and edge-computing architecture [6].

3.12.3 CONSTRAINTS

Upload reliability depends on network connectivity, storage availability, file size, image format, and authenticated device communication.

Failed uploads must not create misleading successful capture records.

3.12.4 STANDARDS

Image ingest should use authenticated requests, validated file formats, server-side size limits, and consistent metadata records.

3.12.5 PRIORITY

Critical

3.13 OPERATOR TAKE PICTURE COMMAND

3.13.1 DESCRIPTION

The system shall allow an authorized operator to request an immediate image capture for a selected plant, tray, or capture pose.

The edge device shall execute the capture when possible and return a status or stored image record through the application workflow.

3.13.2 SOURCE

Operator inspection workflow and device-management controls.

3.13.3 CONSTRAINTS

Immediate capture depends on device availability, safe motion state, camera readiness, travel limits, and network connectivity.

If a device is offline, the system must report that the command cannot be completed at that time.

3.13.4 STANDARDS

Operator commands should be authenticated, logged, and associated with a clear success, failure, or pending status.

3.13.5 PRIORITY

Critical

3.14 AUTOMATIC PLANT RECORD UPDATE AND DISEASE DETECTION AFTER CAPTURE

3.14.1 DESCRIPTION

After a successful capture, the system shall associate the image with the correct plant or tray record and make the capture available for optional plant-health analysis.

If an approved computer-vision model is configured, the system should store the model result and confidence information with the capture history.

3.14.2 SOURCE

Plant-history workflow, image-based monitoring research, plant-health image datasets, and optional computer-vision functionality [1, 2, 4].

3.14.3 CONSTRAINTS

Model output depends on dataset quality, image quality, lighting, plant species, and model limitations.

The system shall present model output as decision support rather than guaranteed diagnosis.

3.14.4 STANDARDS

Image-analysis behavior should preserve traceability between image, plant, model version if available, confidence value, and displayed result.

3.14.5 PRIORITY

High

3.15 EDGE DEVICE MANAGEMENT UI, CAPTURE POSES & SCHEDULES

3.15.1 DESCRIPTION

The system shall provide Vision Console controls for managing edge devices, linked trays, capture poses, and capture schedules.

Operators shall be able to see device status, initiate capture commands, and update schedule or pose information used by the rack-mounted camera carriage.

3.15.2 SOURCE

Operator workflow, rack-scanning system design, current hardware integration, and edge-computing principles [6].

3.15.3 CONSTRAINTS

UI controls must respect authentication and operator authorization.

Pose and schedule values must remain within the physical travel limits and safe operating assumptions of the rack-mounted camera system.

3.15.4 STANDARDS

Device-management workflows should follow standard authenticated configuration, input validation, and status-display practices.

3.15.5 PRIORITY

Critical

4 PACKAGING REQUIREMENTS

This section defines how AgriHome shall be prepared, assembled, and delivered for demonstration and project review.

Packaging requirements cover the physical prototype, software source, configuration, documentation, and setup materials needed to reproduce or evaluate the system.

4.1 HARDWARE PACKAGING

4.1.1 DESCRIPTION

The hardware prototype shall include the plant rack, camera module, rack-mounted camera carriage, X-axis and Y-axis motion components, Raspberry Pi-class edge device, required wiring, mounting fixtures, and safe cable routing.

Hardware shall be packaged or arranged so the system can be transported, stored, and demonstrated without damaging the camera, plants, rails, wiring, or rack structure.

4.1.2 SOURCE

Project scope, physical design, and demonstration needs.

4.1.3 CONSTRAINTS

Packaging is constrained by available budget, lab access, rack dimensions, fixture materials, safe motion limits, and transportability.

4.1.4 STANDARDS

Hardware packaging should follow safe lab-handling practices and manufacturer recommendations for powered electronics, camera hardware, and motion components.

4.1.5 PRIORITY

High

4.2 SOFTWARE PACKAGING

4.2.1 DESCRIPTION

The software package shall include the Vision Console application, application services, database or storage setup instructions, edge-device software, environment configuration examples, and documentation needed to run the prototype.

4.2.2 SOURCE

Project deliverables, maintainability needs, and demonstration setup.

4.2.3 CONSTRAINTS

Some deployment values, credentials, device keys, and storage paths must not be committed publicly.

Setup documentation must distinguish development, demonstration, and device configuration.

4.2.4 STANDARDS

Software packaging should follow common repository organization, README, environment-variable, and deployment documentation practices.

4.2.5 PRIORITY

High

5 PERFORMANCE REQUIREMENTS

This section defines measurable or observable performance expectations for the AgriHome prototype.

Performance requirements focus on reliability, response time, image handling, and practical operation under senior-design demonstration conditions.

5.1 RELIABILITY

5.1.1 DESCRIPTION

The system shall reliably support normal monitoring workflows during demonstration: operator login, device status review, capture request, image ingest, capture-history display, and plant-status review.

5.1.2 SOURCE

Sponsor expectations, product concept, and final demonstration workflow.

5.1.3 CONSTRAINTS

Reliability depends on power, network stability, image-storage availability, device readiness, rack alignment, and camera motion.

Prototype reliability may be lower than production reliability.

5.1.4 STANDARDS

The system should log or display failure states clearly enough for operators and developers to diagnose failed captures or offline devices.

5.1.5 PRIORITY

Critical

5.2 SPEED

5.2.1 DESCRIPTION

The Vision Console should load common dashboard, plant, tray, device, and capture-history views within a practical response time during demonstration.

On-demand capture should return an understandable status promptly, even when the physical capture takes longer to complete.

5.2.2 SOURCE

Operator workflow and usability expectations.

5.2.3 CONSTRAINTS

Response time depends on local network conditions, server availability, image size, storage location, database performance, and edge-device activity.

5.2.4 STANDARDS

The user interface should follow common web responsiveness practices and provide visible feedback for long-running capture operations.

5.2.5 PRIORITY

High

5.3 IMAGE QUALITY

5.3.1 DESCRIPTION

Captured images shall be clear enough for a human operator to identify the target plant and inspect visible leaf or growth changes.

When computer vision is used, captured images should be consistent enough to support basic model evaluation.

5.3.2 SOURCE

Plant-monitoring requirements and research on plant imaging, disease detection, datasets, and phenotyping [1, 2, 3].

5.3.3 CONSTRAINTS

Image quality depends on lighting, focus, camera placement, plant movement, rack vibration, lens cleanliness, and capture distance.

5.3.4 STANDARDS

Image capture should use consistent lighting and documented capture pose assumptions where practical.

5.3.5 PRIORITY

High

6 SAFETY REQUIREMENTS

This section defines safety requirements for operating the AgriHome hardware prototype.

Safety requirements are important because the current design includes powered electronics, wiring, a moving camera carriage, and a physical rack used near users and plants.

6.1 LABORATORY EQUIPMENT LOCKOUT/TAGOUT PROCEDURES

6.1.1 DESCRIPTION

Before maintenance, wiring changes, mechanical adjustments, or motion-system repair, the team shall power down relevant hardware and prevent accidental activation of the moving camera carriage.

6.1.2 SOURCE

University lab safety practices and powered-equipment handling.

6.1.3 CONSTRAINTS

The prototype may not include industrial lockout equipment, but the team must apply a practical student-lab equivalent: disconnect power, communicate maintenance status, and keep operators away from the moving carriage.

6.1.4 STANDARDS

General laboratory equipment safety and lockout/tagout awareness practices.

6.1.5 PRIORITY

Critical

6.2 ELECTRICAL WIRING COMPLIANCE

6.2.1 DESCRIPTION

The system shall use safe low-voltage wiring practices for the Raspberry Pi edge device, camera, motor drivers, and related powered components.

Wires shall be routed to avoid pinching, snagging, overheating, or interfering with moving parts.

6.2.2 SOURCE

Hardware assembly requirements and lab safety expectations.

6.2.3 CONSTRAINTS

The prototype is constrained by available parts, student fabrication resources, and the need to keep wiring accessible for testing.

6.2.4 STANDARDS

Applicable low-voltage electronics practices, manufacturer instructions, and university lab safety guidance.

6.2.5 PRIORITY

Critical

6.3 MOVING CARRIAGE SAFETY

6.3.1 DESCRIPTION

The rack-mounted camera carriage shall move within defined travel limits and shall not intentionally contact plants, rack supports, wiring, or users.

Operators shall have a clear way to stop or power down the system if motion becomes unsafe.

6.3.2 SOURCE

Rack-mounted XY motion design and physical prototype risk review.

6.3.3 CONSTRAINTS

Travel limits, stepper behavior, mechanical alignment, rack rigidity, and wiring slack affect safe motion.

6.3.4 STANDARDS

Safe prototype motion practices for student laboratory demonstrations.

6.3.5 PRIORITY

Critical

7 SECURITY REQUIREMENTS

This section defines security requirements for AgriHome.

Security requirements protect operator access, device communication, plant-image records, and system configuration.

7.1 USER AUTHENTICATION

7.1.1 DESCRIPTION

The system shall require authorized users to authenticate before accessing the Vision Console, plant records, device-management controls, capture history, or configuration settings.

7.1.2 SOURCE

Operator access control and project security expectations.

7.1.3 CONSTRAINTS

Authentication implementation may depend on the selected application framework and available university or project infrastructure.

7.1.4 STANDARDS

Common web authentication and session-management practices.

7.1.5 PRIORITY

Critical

7.2 DATA ENCRYPTION

7.2.1 DESCRIPTION

The system should use encrypted communication for browser-to-application traffic and should avoid transmitting credentials, device keys, or private configuration values in plaintext across untrusted networks.

7.2.2 SOURCE

Secure communication expectations and device-management requirements.

7.2.3 CONSTRAINTS

Local demonstration networks may limit certificate setup, but the production-oriented design should assume encrypted communication.

7.2.4 STANDARDS

Common HTTPS and secure API communication practices.

7.2.5 PRIORITY

High

7.3 SECURE DATA STORAGE

7.3.1 DESCRIPTION

The system shall store plant records, capture metadata, image references, and operator-access data in a controlled storage environment.

Sensitive credentials and device keys shall not be exposed in public documentation or client-side code.

7.3.2 SOURCE

Data-management requirements and project security expectations.

7.3.3 CONSTRAINTS

Prototype storage may use available teammate or university resources, but access should remain limited to authorized project members and operators.

7.3.4 STANDARDS

Common database access-control, secret-management, and data-retention practices.

7.3.5 PRIORITY

High

7.4 DEVICE API KEY SECURITY

7.4.1 DESCRIPTION

Registered edge devices shall use a device-specific credential or key when communicating with application services.

The application shall treat unknown or revoked device credentials as unauthorized.

7.4.2 SOURCE

Edge-device integration, image-ingest security requirements, and distributed edge-computing concepts [6].

7.4.3 CONSTRAINTS

Device credentials must be provisioned, stored, rotated, and protected carefully enough for prototype operation without overcomplicating demonstration setup.

7.4.4 STANDARDS

Common API-key management, credential rotation, and least-privilege communication practices.

7.4.5 PRIORITY

Critical

8 MAINTENANCE & SUPPORT REQUIREMENTS

This section defines requirements for maintaining and supporting the AgriHome prototype during development, demonstration, and project handoff.

8.1 DEVICE RECOVERY

8.1.1 DESCRIPTION

The system shall support practical recovery when the Raspberry Pi edge device, camera, network connection, or application service fails.

Recovery documentation shall describe restart steps, basic troubleshooting, and how operators can identify stale or offline devices.

8.1.2 SOURCE

Operational reliability needs and demonstration support.

8.1.3 CONSTRAINTS

Recovery procedures are constrained by available hardware access, network conditions, and the prototype nature of the system.

8.1.4 STANDARDS

Common troubleshooting documentation and operational support practices.

8.1.5 PRIORITY

High

8.2 IMAGE STORAGE MAINTENANCE

8.2.1 DESCRIPTION

The system shall include a maintainable approach for storing captured images and related metadata.

The team shall document storage locations, retention expectations, and how to remove or archive test images when needed.

8.2.2 SOURCE

Image-history requirement and storage-management needs.

8.2.3 CONSTRAINTS

Image volume may grow quickly depending on capture frequency, image resolution, and number of plants.

8.2.4 STANDARDS

Common file-storage, backup, cleanup, and metadata-management practices.

8.2.5 PRIORITY

High

8.3 DOCUMENTATION AND SUPPORT

8.3.1 DESCRIPTION

The project shall provide documentation for installing, configuring, operating, troubleshooting, and demonstrating the AgriHome system.

Documentation shall cover the Vision Console, plant and tray management, edge-device setup, capture workflows, image history, and safe motion operation.

8.3.2 SOURCE

Course deliverables, project handoff needs, and operator-support expectations.

8.3.3 CONSTRAINTS

Documentation must remain accurate as implementation details change and must not expose secrets or private deployment values.

8.3.4 STANDARDS

Common README, setup guide, user manual, and troubleshooting documentation practices.

8.3.5 PRIORITY

High

9 OTHER REQUIREMENTS

This section contains requirements that do not fit cleanly into the preceding categories but still affect the AgriHome prototype and project delivery.

9.1 BUDGET ALIGNMENT

9.1.1 DESCRIPTION

The system shall remain within the approved project budget unless additional funding is formally approved. Hardware and materials choices should support the current rack-mounted camera design without adding unnecessary cost.

9.1.2 SOURCE

Project charter and sponsor funding constraints.

9.1.3 CONSTRAINTS

Part availability, shipping time, university-provided materials, and 3D-print access may affect final cost.

9.1.4 STANDARDS

Course project budgeting and cost-tracking expectations.

9.1.5 PRIORITY

High

9.2 COURSE DEMONSTRATION READINESS

9.2.1 DESCRIPTION

The system shall be prepared for senior-design demonstration with enough reliability, documentation, and setup clarity that the team can show the rack-mounted capture workflow, device status, image ingest, capture history, and Vision Console behavior.

9.2.2 SOURCE

Course deliverables and final project review expectations.

9.2.3 CONSTRAINTS

Demonstration readiness depends on hardware availability, network access, stable power, prepared plants, and working application services.

9.2.4 STANDARDS

UTA senior-design deliverable and demonstration expectations.

9.2.5 PRIORITY

Critical

10 FUTURE ITEMS

This section identifies potential future enhancements that are outside the minimum required prototype but may improve AgriHome after the current course project.

10.1 EXPANDED COMPUTER VISION

10.1.1 DESCRIPTION

Future versions should support additional plant species, disease categories, segmentation behavior, model comparison, and model-evaluation reporting.

10.1.2 SOURCE

Plant disease detection, plant-health image datasets, agricultural deep learning, and product-expansion goals [1, 2, 4].

10.1.3 CONSTRAINTS

Additional model capability requires more labeled images, validation data, training time, and careful explanation of model limitations.

10.1.4 STANDARDS

Computer-vision evaluation should document datasets, accuracy limits, confidence behavior, and known failure cases.

10.1.5 PRIORITY

Future

10.2 REMOTE NOTIFICATIONS

10.2.1 DESCRIPTION

Future versions should send remote alerts when abnormal plant status, failed captures, or stale device heartbeats require attention.

10.2.2 SOURCE

Customer convenience and maintenance workflow.

10.2.3 CONSTRAINTS

Notification support requires additional accounts, opt-in settings, backend services, secure tokens, and delivery testing.

10.2.4 STANDARDS

Secure notification and user-consent practices.

10.2.5 PRIORITY

Future

10.3 ENVIRONMENTAL SENSOR EXPANSION

10.3.1 DESCRIPTION

Future versions should integrate environmental sensors for temperature, humidity, light level, and soil moisture so image history can be interpreted alongside growing conditions.

10.3.2 SOURCE

IoT and edge-computing literature supporting connected monitoring systems [5, 6].

10.3.3 CONSTRAINTS

Sensor expansion requires additional hardware, calibration, storage fields, UI design, and validation work.

10.3.4 STANDARDS

Manufacturer sensor specifications and controlled-environment monitoring practices.

10.3.5 PRIORITY

Future

10.4 MULTI-RACK SUPPORT

10.4.1 DESCRIPTION

Future versions should support multiple racks, multiple edge devices, and a larger number of plants while preserving clear device status, image history, and operator workflows.

10.4.2 SOURCE

Product expansion, distributed IoT architecture, and edge-device scalability concepts [5, 6].

10.4.3 CONSTRAINTS

Multi-rack operation increases scheduling complexity, storage usage, device-management needs, and hardware cost.

10.4.4 STANDARDS

Common scalable device-management and data-organization practices.

10.4.5 PRIORITY

Future

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